IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-334

1. Topic (12 words max)

Athlete Performance and Training AI-Based System

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Athletic performance optimization has traditionally relied on coaches' experience, manual observation, and post-training analysis. However, this approach often lacks precision, personalization, and predictive insight. As athletes face increasing pressure to perform while avoiding injury, there is a growing need for intelligent systems that can monitor, analyze, and improve training effectiveness in real-time.

Modern sports generate vast amounts of data from wearable devices, video analysis, and biometric monitoring, yet much of this data remains underutilized. Key performance indicators—such as heart rate variability, training load, sleep patterns, and form technique—can provide valuable insights if analyzed effectively. However, manually processing this data is time-consuming and error-prone. This creates a gap where AI and machine learning can offer significant value.

The proposed research aims to address this gap by developing an AI-based system that can collect and process diverse training and physiological data to enhance athlete performance. The system will leverage wearable technology, pose estimation algorithms, and machine learning models to perform tasks such as predicting performance outcomes, identifying injury risks, analyzing movement technique, and recommending personalized training plans.

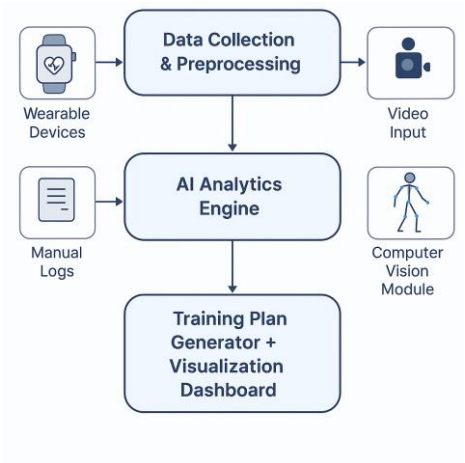
This research not only contributes to the field of sports science and artificial intelligence but also has practical implications for coaches, physiotherapists, and athletes seeking data-driven support. The outcome will be a functional prototype backed by experimental results and real-world use cases to demonstrate its effectiveness and feasibility.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is an **AI-based system** designed to monitor, predict, and enhance athlete performance through the integration of wearable data, , and video analysis. The system is composed of four core modules:

1. **Data Acquisition Module**: Collects real-time data from wearable sensors (e.g., heart rate, distance, GPS), user inputs (nutrition, sleep), and video feeds for motion analysis.
2. **AI Analytics Engine**: Uses machine learning models to predict performance trends, detect injury risks, and identify anomalies in biometric data. It includes:
    - Regression models for performance prediction
    - Classification models for injury detection
    - Clustering models for athlete profiling
3. **Computer Vision Module**: Utilizes pose estimation tools (like OpenPose or MediaPipe) to evaluate athlete form and technique from video data, identifying posture-related inefficiencies or potential strain patterns.
4. **Recommendation & Visualization Layer**: Generates personalized training plans using reinforcement learning and displays insights through an interactive dashboard, showing metrics, trends, and alerts.

This modular system allows seamless data flow, from input to actionable insight, helping athletes and coaches make informed decisions. The novelty lies in its **multimodal integration** and **real-time feedback capability**, enabling a comprehensive, adaptive training approach.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The development of an AI-based athlete performance and training system requires **interdisciplinary expertise** that spans sports science, artificial intelligence, biomechanics, and software engineering.

Specialized Domain Expertise & Knowledge:

1. **Sports Science & Physiology**
   - Understanding of key athletic performance indicators (VO2 max, heart rate variability, fatigue, training load).
   - Knowledge of injury patterns, overtraining symptoms, and biomechanics to guide model development and validation.
2. **Machine Learning & AI**
   - Proficiency in supervised and unsupervised learning algorithms (e.g., regression, classification, clustering).
   - Experience in training predictive models (e.g., Random Forest, XGBoost), pose analysis with computer vision, and reinforcement learning for adaptive systems.
3. **Computer Vision & Pose Estimation**
   - Use of tools like OpenPose or MediaPipe for detecting and evaluating human posture and movement.
   - Understanding video processing, model inference, and temporal motion tracking.
4. **Full-Stack Development**
   - Backend knowledge in data processing (Flask/FastAPI), model integration, and database management.
   - Frontend skills to develop real-time dashboards using frameworks like React.js or Flutter.

Data Requirements:

1. **Wearable Sensor Data**
   - Heart rate, GPS, pace, calorie burn, sleep metrics—collected via APIs (Fitbit, Garmin, etc.)
2. **Video Data**
   - Sports footage for technique analysis, preferably annotated or labeled.
3. **Public Datasets**
   - Datasets from Kaggle (e.g., athlete training logs, Olympic performance), PoseTrack, UCF101, etc.

8. Objectives and Novelty

Main Objective

To design and develop an AI-powered system that enhances athlete performance by collecting, analyzing, and interpreting multimodal training and data, enabling predictive insights, injury prevention, technique correction, and personalized training recommendations.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Y.Manulakshan IT22283962	Develop a robust data acquisition and integration system	• Integrate wearable device APIs (Fitbit, Garmin, etc.) • Set up video capture for form and posture tracking • Build and maintain database structure for centralized data storage • Perform data preprocessing and normalization	• Unified system for **multi-modal data collection** (video, manual input) • Real-time integration from various athlete monitoring sources • Customized data preprocessing pipelines optimized for athletic metrics

L.M.Sylvester IT22311368	Design and train AI models for performance prediction	• Build regression models for performance forecasting • Develop classification models to assess injury/fatigue risk • Implement anomaly detection for unusual biometrics or patterns • Evaluate models using public/simulated athlete datasets • Optimize models through hyperparameter tuning and cross-validation	• AI-powered system that can **proactively predict injury risk** • Use of **multi-dimensional data (training, biometrics, sleep)** to enhance model accuracy
G.Jude Jawakker IT22330864	Build a personalized, intelligent training and visualization system	• Develop frontend dashboard (React.js / Next.js) • Implement reinforcement learning or decision trees for adaptive training plans • Integrate all modules into a full-stack prototype • Deploy and test final prototype with sample users or feedback loop	• AI-powered **adaptive training recommender system** • Use of **Reinforcement Learning** in athlete performance optimization • Unified interface for real-time feedback and **insight-driven training adjustments**

G.Mathumitha IT21379338	Implement computer vision for form and technique analysis	• Set up pose estimation tools (OpenPose, MediaPipe) • Analyze posture and form using video datasets • Visualize form feedback for athletes in the dashboard • Annotate or augment datasets for supervised learning	• Automated technique assessment using **real-time pose estimation** • Form error detection applied to athletic performance • Integration of **computer vision and biomechanical analysis** in training systems

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Y.Manulakshan IT22283962	Involves API integration, database design, and secure data handling—all key areas in software engineering and data management. • **Technologies**: REST APIs (e.g., Fitbit), Firebase/MongoDB/PostgreSQL for data storage, Python for preprocessing. • **IT Concepts**: Data pipelines, backend development, real-time data synchronization, and normalization techniques.
L.M.Sylvester IT22311368	Focuses on artificial intelligence and machine learning, central domains within modern IT. • **Technologies**: Python (scikit-learn, TensorFlow, XGBoost), Jupyter for prototyping, classification/regression algorithms. • **IT Concepts**: Predictive modeling, algorithm development, anomaly detection, and performance evaluation.
G.Jude Jawakker IT22330864	Involves full-stack development, UX/UI design, and intelligent system recommendations. • **Technologies**: React.js or Next js Web (frontend), Flask/FastAPI (backend), reinforcement learning for decision-making. • **IT Concepts**: Web application development, adaptive systems, data visualization, API integration.
G.Mathumitha IT21379338	Applies advanced IT concepts such as computer vision, real-time video analysis, and pose estimation. • **Technologies**: OpenPose / MediaPipe for skeleton tracking, Python for frame processing. • **IT Concepts**: Image processing, human-computer interaction, AI-based motion analysis, and CV model integration.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Professor	Samantha	Rajapaksha	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.